

MD. ROFAZ HASAN RAFIU

Machine Learning • Computer Vision • Applied AI Engineer

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PROFESSIONAL SUMMARY

Machine Learning & Applied Computer Vision Engineer with a strong Computer Science foundation at RUET. Specialized in edge computer vision, in-browser WebAssembly model execution (OpenCV.js, TensorFlow.js), document image preprocessing pipelines, and relational ML data infrastructure. Focused on building zero-infrastructure-cost applied ML solutions running client-side with deterministic fallbacks and scalable backend data pipelines.

EDUCATION

Rajshahi University of Engineering & Technology (RUET) — B.Sc. in Computer Science & Engineering 2023 – 2027

- Admitted through national combined engineering entrance examination (CUET, KUET, RUET) ranking in top merit percentile.
- Relevant Coursework: Algorithms & Data Structures, Linear Algebra, Probability & Statistics, Discrete Mathematics, Database Systems.

St. Joseph Higher Secondary School — Higher Secondary Certificate (HSC), Science 2022

- GPA: 5.00 / 5.00

TECHNICAL SKILLS

Computer Vision & Edge ML: OpenCV.js (WebAssembly), TensorFlow.js (BlazeFace, WebGL), Image Preprocessing (Adaptive Thresholding, Canny Edge Detection, Contour Analysis, 4-Point Homography, Perspective Warp, ROI Segmentation), Tesseract.js (OCR), Pixel Histogram Classification.

Applied AI & Local Inference: Local SLM Deployment (Ollama / Qwen 2.5), Structured Output Extraction, Error Classification Heuristics, Sub-10ms Rule-Based Fallback Routines, Confidence Calibration.

Languages & Frameworks: Python (NumPy, OpenCV basics, FastAPI), TypeScript, Go 1.22, C++, React 19, Next.js 15, Flutter (Dart).

Data & Systems Engineering: PostgreSQL (Prisma ORM, Indexing, Connection Pooling), Redis, SQLite, Docker, Linux/Bash, RESTful APIs, Git, GitHub Actions, Jest.

FEATURED MACHINE LEARNING & VISION SYSTEMS

OMRView — In-Browser Computer Vision Processing Pipeline [TypeScript, React 19, Vite, OpenCV.js Wasm] 2025

Client-side optical mark recognition and document rectification engine | Code: github.com/rofazhasan/OMRView

- Engineered a fully client-side computer vision pipeline utilizing @techstark/opencv-js WebAssembly build, eliminating all backend GPU/server compute costs for document processing.
- Implemented a multistage image transformation sequence: adaptive thresholding, Canny edge detection, Douglas-Peucker polygon approximation, and 4-point homography warp to correct skewed camera captures.
- Designed a bubble fill classification algorithm calculating relative pixel intensity histograms across segmented Regions of Interest (ROIs), achieving resilience across varying lighting and document degradation.
- Architected responsive React 19 UI with real-time visual overlays rendering rectified document contours and instantaneous question mark verification.

Digital School — Vision Monitoring & SLM Mistake Diagnostic Engine [Next.js 15, TensorFlow.js, Ollama, PostgreSQL] 2024 – Present

Production digital education platform with client vision models and local SLM diagnostics | Live: rofazacademy.dev

- Integrated TensorFlow.js BlazeFace face-detection models with client-side WebGL acceleration to monitor examinee presence and orientation during timed online assessments.
- Constructed a local SLM inference service (Ollama / Qwen 2.5) that analyzes student exam error distributions and categorizes mistakes into concrete academic concept weaknesses.
- Engineered deterministic sub-10ms fallback routines that compute accurate topic error metrics if local model inference is delayed or unavailable, maintaining 100% assessment uptime.
- Developed multi-tenant PostgreSQL data models in Prisma with optimized indexing and connection pooling for high-concurrency exam submissions and automated grading.

Krishi Bondhu AI — Crop Disease Mobile Vision Assistant [Flutter, Dart, Riverpod, sqflite, Hardware Camera API] 2024

Offline-first edge mobile image collection and diagnostic application | Code: github.com/rofazhasan/Krishi_Bondhu_AI

- Built an edge mobile image acquisition pipeline capturing and standardizing crop leaf images directly from device camera hardware with real-time framing guides.
- Implemented an offline-first architecture with local SQLite storage, allowing field workers in low-connectivity agricultural areas to record and queue diagnostic data without network dependency.

FocusGuard — Distributed Systems Daemon & WebSocket Agent [Go 1.22, WebSockets, Docker, Chrome MV3] 2025

High-concurrency systems infrastructure for distributed client state tracking | Code: github.com/rofazhasan/FocusGuard

- Engineered a high-performance Go daemon tracking client activity and broadcasting synchronization events over persistent low-latency WebSockets.
- Containerized multi-service environment with Docker Compose and implemented GitHub Actions CI for automated build verification.

BioEntry Pro v6.0 — Universal Biometric Verification System [Electron, React, TensorFlow.js, WebGL, WASM, Node.js] 2024

Cross-platform biometric attendance system with adaptive hardware acceleration | Code: github.com/rofazhasan/BioEntry

- Built an enterprise biometric attendance engine with TensorFlow.js (@vladmandic/human) supporting real-time face detection and multi-camera stream processing.
- Implemented dynamic hardware acceleration fallback (WebGL → WASM → CPU) and automated binary bundling for universal Apple Silicon & x64 architectures.

Atlas — macOS Native Operating System & Document OCR [Swift 6, SwiftUI, SwiftData, Apple Vision OCR, CryptoKit]2024

Native macOS financial tracking and document intelligence | Code: github.com/rofazhasan/atlas

- Integrated Apple Vision framework for offline tabular receipt scanning and OCR parsing into SwiftData models with AES-256 GCM encrypted storage.

Additional Open-Source Systems & Repositories

GitHub: github.com/rofazhasan

- **Study Companion** (github.com/rofazhasan/study-companion): AI learning companion built with Flutter, Dart, Firebase, and Google Gemini AI.
- **Meal Manager V2** (github.com/rofazhasan/Meal-Management): Fullstack relational meal calculation and mess expense ledger in React 18, TypeScript, and Prisma v5.
- **sdms** (github.com/rofazhasan/sdms): High-concurrency educational administrative platform with student record indexing.
- **Election Game** (github.com/rofazhasan/Election_Game): Political strategy simulation engine and finite state machine in Godot Engine / GDScript.

ACHIEVEMENTS & LEADERSHIP

Creative Talent Hunt — Regional Champion (Mathematics & Computing)

Government of Bangladesh

Selected among top student problem solvers across the division in national competitive talent assessment.

RUET CSE Peer Workshop Instructor & Competitive Programmer

2023 – Present

Conducted technical mentoring sessions on Data Structures, Algorithms, and Object-Oriented Architecture; active participant in university programming contests.